

## **Task 2: Embedded Systems Lab Setup & Sensor Interfacing**

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### **1. Objective**

The objective of this task is to set up a functional virtual embedded development environment using Tinkercad and gain practical experience interfacing a temperature sensor with a microcontroller. By building this project, I am writing basic embedded software logic to handle real-time sensor data and trigger automation responses based on predefined thresholds.

### **2. Selected Working Environment**

- **Platform:** Tinkercad Circuits Simulator
- **Sensor Choice:** TMP36 Temperature Sensor (Analog)
- **Actuators:** LED (Pin 13) & Piezo Buzzer (Pin 12)

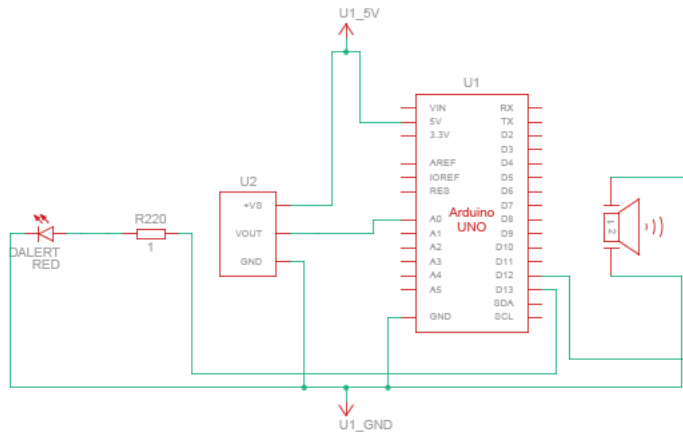
### **3. Hardware Interfacing & Circuit Diagram**

The TMP36 is an analog sensor that requires connection to an Analog-to-Digital Converter (ADC) pin on the Arduino Uno.

#### **Connection Mapping:**

- **TMP36 Pin 1 (Power):** Connected to Arduino **5V** power rail.
- **TMP36 Pin 2 (Vout):** Connected to Arduino **Analog Pin A0**.
- **TMP36 Pin 3 (GND):** Connected to Arduino **GND** (Ground).
- **LED (Anode / Long Leg):** Connected to Arduino **Digital Pin 13** via a 220 Ohm current-limiting resistor.
- **LED (Cathode / Short Leg):** Connected to Arduino **GND**.
- **Buzzer Positive Pin (+):** Connected to Arduino **Digital Pin 12**.
- **Buzzer Negative Pin (-):** Connected to Arduino **GND**.

### Circuit diagram:



### 4. Source Code (.ino)

The following program initializes the serial monitor, continuously samples the analog voltage from the TMP36 sensor, converts the reading into accurate Celsius degrees, and executes a threshold alert logic to activate the LED and buzzer.

```
// Task 2: Embedded Systems & IOT - TMP36 Sensor Interfacing (Tinkercad)
```

```
// Define Hardware Pins
```

```
const int sensorPin = A0; // TMP36 Vout pin connected to A0
```

```
const int ledPin = 13; // Alert LED connected to Pin 13
```

```
const int buzzerPin = 12; // Alert Buzzer connected to Pin 12
```

```
// Alert Threshold Configuration
```

```
const float thresholdTemp = 30.0; // Alert triggers if temp > 30°C
```

```
void setup()
```

```
{
```

```

// Initialize Serial Communication at 9600 bps
Serial.begin(9600);

// Configure Actuator Pins as Outputs
pinMode(ledPin, OUTPUT);
pinMode(buzzerPin, OUTPUT);
}

void loop()
{
    // 1. Read raw analog integer value from TMP36 (0 to 1023)
    int rawValue = analogRead(sensorPin);

    // 2. Convert raw data to voltage (0V - 5V) based on 10-bit resolution
    float voltage = rawValue * (5.0 / 1023.0);

    // 3. Convert voltage to Temperature in Celsius
    // TMP36 has a 500mV (0.5V) offset. Scale factor is 10mV per °C.
    float temperatureC = (voltage - 0.5) * 100.0;

    // 4. Output the results to the Serial Monitor
    Serial.print("Temperature: ");
    Serial.print(temperatureC);
    Serial.println(" °C");

    // 5. Threshold checking logic
    if (temperatureC > thresholdTemp)
    {
        digitalWrite(ledPin, HIGH); // Turn LED ON
        digitalWrite(buzzerPin, HIGH); // Turn Buzzer ON

        Serial.println("🚨 ALERT: High Temperature Threshold Exceeded!");
    }
    else
    {

```

```

digitalWrite(ledPin, LOW);    // Turn LED OFF
digitalWrite(buzzerPin, LOW); // Turn Buzzer OFF
}

// Delay for 2 seconds before the next reading
delay(2000);
}

```

## 5. Functional Explanation

### How the TMP36 Sensor Works:

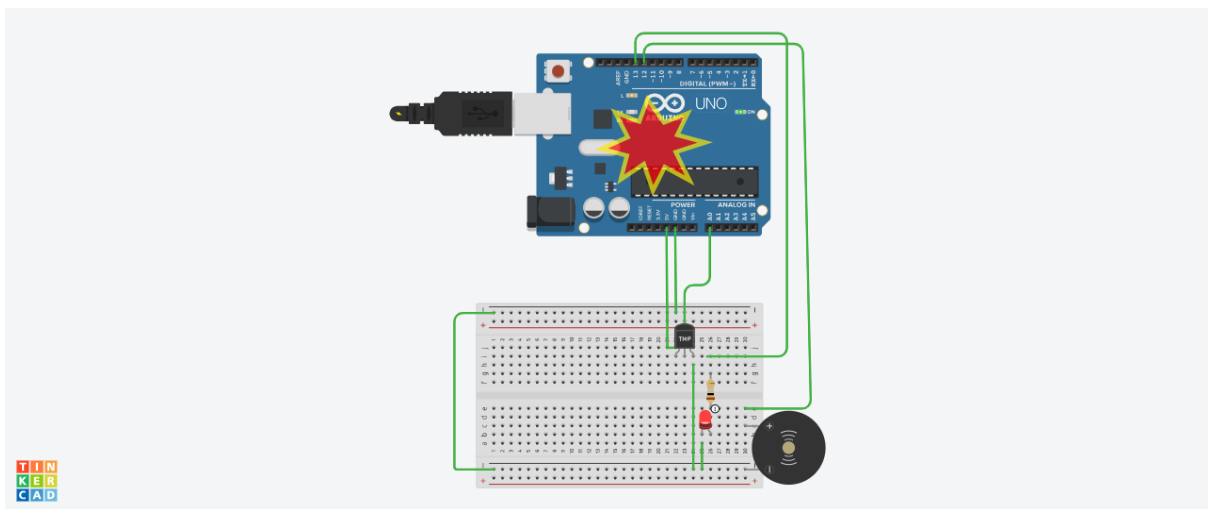
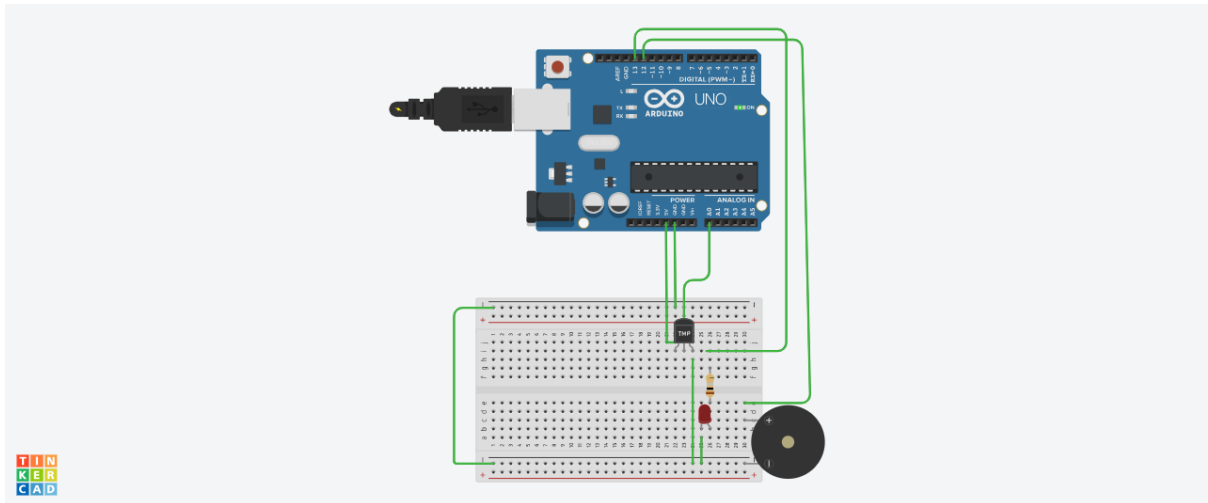
The TMP36 is a precision integrated-circuit temperature sensor whose output voltage varies linearly with the Celsius temperature scale.

### What the Code Does:

1. **Data Acquisition:** The Arduino uses its internal 10-bit ADC to sample the voltage on pin A0. This maps the input voltage into an integer range from 0 to 1023.
2. **Mathematical Transformation:** The formula  $\text{rawValue} * (5.0 / 1023.0)$  converts the raw integer back into a real voltage value.
3. **Offset and Temperature Scaling:** The code subtracts 0.5 to remove the sensor's offset and multiplies the remainder by 100 to yield the precise temperature variable (temperatureC).
4. **Automation Response:** The code prints data values to the Serial Monitor every 2 seconds. An if-else block continually checks the calculated temperature against the configured thresholdTemp. If the sensor reads higher than , the system writes a HIGH signal to digital pins 12 and 13, illuminating the alert LED, activating the buzzer, and broadcasting a warning over the serial bus.

## 6. Verification & Outcomes

- **Serial Monitor Verification:**



### Key Achievements:

- Developed a fully functional virtual embedded testing environment in Tinkercad.
- Configured analog data mapping and offset compensation formulas in C++.
- Established an automatic multi-actuator threshold alert layout, laying the framework for complex IoT automation systems.